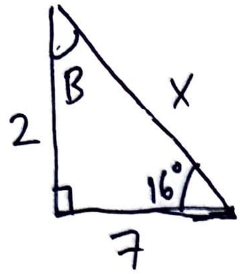
Final Exam Review

KEY

Unit 4A

1.



$\sin 16° = \frac{2}{X} \Rightarrow X = \frac{2}{\sin 16°} = 2\csc 16°$

$\angle B = 180° - 90° - 16°$

$\boxed{X = 2\csc 16° \sim 7.26 \quad \angle B = 74°}$

2. a) $\cos\frac{\pi}{3} = \boxed{\frac{1}{2}}$ b) $\tan\left(\frac{5\pi}{6}\right) = \boxed{\frac{-\sqrt{3}}{3}}$ c) $\sin(\pi) = \boxed{0}$

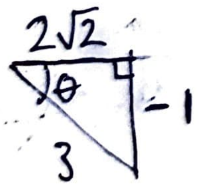
d) $\csc\left(\frac{2\pi}{3}\right) = \boxed{\frac{2\sqrt{3}}{3}}$ e) $\sec\left(\frac{5\pi}{4}\right) = \boxed{-\sqrt{2}}$ f) $\cot\left(\frac{7\pi}{3}\right) = \boxed{\frac{\sqrt{3}}{3}}$

$= \frac{1}{\sin(2\pi/3)} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

$= \frac{1}{\cos(5\pi/4)} = \frac{1}{-\sqrt{2}/2} = \frac{-2}{\sqrt{2}}$

$\frac{7\pi}{3}$ co-term w/ $\frac{\pi}{3}$

3. Given $\sin\theta = \frac{-1}{3}$ and $\theta \in Q\,\text{IV}$



$\sqrt{3^2 - 1^2}$
$= \sqrt{8}$
$= 2\sqrt{2}$

a) $\cos\theta = \boxed{\frac{2\sqrt{2}}{3}}$

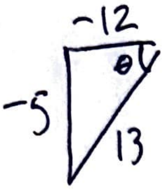
b) $\tan\theta = \frac{-1}{2\sqrt{2}} = \boxed{\frac{-\sqrt{2}}{4}}$

c) $\sec\theta = \frac{1}{\cos\theta} = \frac{3}{2\sqrt{2}} = \boxed{\frac{3\sqrt{2}}{4}}$

d) $\csc\theta = \frac{1}{\sin\theta} = \boxed{-3}$

e) $\cot\theta = \frac{1}{\tan\theta} = \frac{-4}{\sqrt{2}} = \boxed{-2\sqrt{2}}$

4. Given $\tan\theta = \frac{5}{12}$ and $\theta \in Q\,III$



a) $\sin\theta = \boxed{-5/13}$

b) $\cos\theta = \boxed{-12/13}$

c) $\sec\theta = \boxed{-13/12}$

d) $\csc\theta = \boxed{-13/5}$

e) $\cot\theta = \boxed{5/12}$

5. Solve for $0 \le \theta < 2\pi$

a) $2\sin\theta + 1 = 0$

$\Rightarrow \sin\theta = -1/2$

$\boxed{\therefore \theta = \left\{\frac{7\pi}{6}, \frac{11\pi}{6}\right\}}$

b) $\tan^2\theta - 3 = 0$

$\tan^2\theta = 3$

$\tan\theta = \pm\sqrt{3}$

$\boxed{\therefore \theta = \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\right\}}$

c) $\sin(3\theta) = -\frac{\sqrt{2}}{2}$

$3\theta = \frac{5\pi}{4} + 2\pi n,\ n \in \mathbb{Z}$

$\Rightarrow \theta = \frac{5\pi}{12} + \frac{2\pi}{3}n = \frac{5\pi}{12} + \frac{8\pi}{12}n,\ n \in \mathbb{Z}$

and

$3\theta = \frac{7\pi}{4} + 2\pi n,\ n \in \mathbb{Z}$

$\Rightarrow \theta = \frac{7\pi}{12} + \frac{8\pi}{12}n,\ n \in \mathbb{Z}$

$\boxed{\therefore \theta = \begin{pmatrix} \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{21\pi}{12} \\ \frac{7\pi}{12}, \frac{15\pi}{12}, \frac{23\pi}{12} \end{pmatrix}}$

d) $\cos^2(2\theta) = \frac{1}{4}$

$\cos(2\theta) = \pm\sqrt{\frac{1}{4}} = \pm\frac{1}{2}$

θ = every angle making $\pm\frac{1}{2}$ $\div 2$ then $+\pi n, n \in \mathbb{Z}$

Ex: $2\theta = \frac{\pi}{3} + 2\pi n,\ n \in \mathbb{Z}$

$\Rightarrow \theta = \frac{\pi}{6} + \frac{6\pi}{6}n,\ n \in \mathbb{Z}$

$\boxed{\therefore \theta = \left\{ \begin{matrix} \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6} \\ \frac{7\pi}{6}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{11\pi}{6} \end{matrix} \right\}}$

6. Solve for $0 \leq \theta < 2\pi$

a) $\sin\theta + \sin\theta \tan\theta = 0$

$\sin\theta(1 + \tan\theta) = 0$

$\therefore \sin\theta = 0$ or $\tan\theta = -1$

$\boxed{\text{Hence, } \theta = \left\{0, \pi, \frac{3\pi}{4}, \frac{7\pi}{4}\right\}}$

b) $2\cos\theta = \cos\theta \csc\theta$

$2\cos\theta - \cos\theta \csc\theta = 0$

$\cos\theta(2 - \csc\theta) = 0$

$\therefore \cos\theta = 0$ or $\csc\theta = 2$

$\Rightarrow \sin\theta = 1/2$

$\boxed{\text{Hence, } \theta = \left\{\frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}\right\}}$

c) $2\cos^2\theta - \cos\theta = 1$

$2\cos^2\theta - \cos\theta - 1 = 0$

$(2\cos\theta + 1)(\cos\theta - 1) = 0$

$\therefore \cos\theta = -\frac{1}{2}$ or $\cos\theta = 1$

$\boxed{\text{Hence, } \theta = \left\{\frac{2\pi}{3}, \frac{4\pi}{3}, 0\right\}}$

d) $\sin^2\theta + \sin\theta \cot\theta = 1$

$\sin^2\theta + \sin\theta \cdot \frac{\cos\theta}{\sin\theta} - 1 = 0$

$(1 - \cos^2\theta) + \cos\theta - 1 = 0$

$-\cos^2\theta + \cos\theta = 0$

$-\cos\theta(\cos\theta - 1) = 0$

$\therefore \cos\theta = 0$ or $\cos\theta = 1$

$\boxed{\text{Hence, } \theta = \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}}$

* $\theta = 0$ is an extraneous solution because $\cot(0)$ is undefined

7. Fully Simplify each trig expression.

a) $\dfrac{\tan\theta + \cot\theta}{\csc\theta} = \dfrac{\tan\theta + \cot\theta}{\frac{1}{\sin\theta}} = \sin\theta\tan\theta + \sin\theta\cot\theta$

$$\Rightarrow \ = \sin\theta \cdot \frac{\sin\theta}{\cos\theta} + \sin\theta \cdot \frac{\cos\theta}{\sin\theta}$$

$$= \frac{\sin^2\theta}{\cos\theta} + \cos\theta$$

$$= \frac{\sin^2\theta}{\cos\theta} + \frac{\cos^2\theta}{\cos\theta}$$

$$= \frac{\sin^2\theta + \cos^2\theta}{\cos\theta}$$

$$= \frac{1}{\cos\theta}$$

$$= \boxed{\sec\theta}$$

7. b) $\sec\theta(\sec\theta - \cos\theta)$

$= \sec^2\theta - 1$

$= \boxed{\tan^2\theta}$

c) $\tan\theta \csc\theta - \tan^2\theta \cos\theta$

$= \frac{\sin\theta}{\cos\theta} \cdot \frac{1}{\sin\theta} - \frac{\sin^2\theta}{\cos^2\theta} \cdot \cos\theta$

$= \frac{1}{\cos\theta} - \frac{\sin^2\theta}{\cos\theta}$

$= \frac{1-\sin^2\theta}{\cos\theta}$

$= \frac{\cos^2\theta}{\cos\theta}$

$= \boxed{\cos\theta}$

d) $\cos\theta \csc\theta (\sec^2\theta - 1)$

$= \cos\theta \cdot \frac{1}{\sin\theta} (\tan^2\theta)$

$= \cot\theta \cdot \tan^2\theta$

$= \frac{1}{\tan\theta} \cdot \tan^2\theta$

$= \boxed{\tan\theta}$

8. Verify each identity.

a) Show that $\sin^2\theta - \cos^2\theta = 2\sin^2\theta - 1$

Start: $\sin^2 - \cos^2\theta$

$= \sin^2\theta - (1 - \sin^2\theta)$

$= \sin^2\theta - 1 + \sin^2\theta$

$= 2\sin^2\theta - 1$ Q.E.D.

b) Show that $\dfrac{\cos^2\theta}{1+\sin\theta} = 1 - \sin\theta$

Start: $\dfrac{\cos^2\theta}{1+\sin\theta}$

$= \dfrac{1-\sin^2\theta}{1+\sin\theta}$

$= \dfrac{(1+\sin\theta)(1-\sin\theta)}{1+\sin\theta}$

$= 1 - \sin\theta$ Q.E.D.

8. c) Show that $\dfrac{\cos\theta}{1+\sin\theta} = \dfrac{1-\sin\theta}{\cos\theta}$

Start: $\dfrac{\cos\theta}{1+\sin\theta}$

$= \dfrac{\cos\theta}{1+\sin\theta} \cdot \dfrac{(1-\sin\theta)}{(1-\sin\theta)}$

$= \dfrac{\cos\theta\,(1-\sin\theta)}{1-\sin^2\theta}$

$= \dfrac{\cos\theta\,(1-\sin\theta)}{\cos^2\theta}$

$= \dfrac{1-\sin\theta}{\cos\theta}$ Q.E.D.

d) Show that $\dfrac{\cos^4\theta-\sin^4\theta}{\cos^2\theta\sin^2\theta} = \csc^2\theta - \sec^2\theta$

Start: $\dfrac{\cos^4\theta-\sin^4\theta}{\cos^2\theta\sin^2\theta}$

$= \dfrac{(\cos^2\theta+\sin^2\theta)(\cos^2\theta-\sin^2\theta)}{\cos^2\theta\sin^2\theta}$

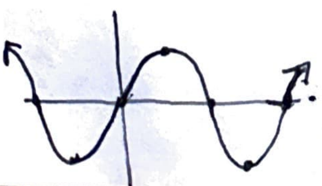
$= \dfrac{\cos^2\theta-\sin^2\theta}{\cos^2\theta\sin^2\theta}$

$= \dfrac{\cos^2\theta}{\cos^2\theta\sin^2\theta} - \dfrac{\sin^2\theta}{\cos^2\theta\sin^2\theta}$

$= \dfrac{1}{\sin^2\theta} - \dfrac{1}{\cos^2\theta}$

$= \csc^2\theta - \sec^2\theta$ Q.E.D.

9. $y = \sin x$



D: $\mathbb{R}$

R: $[-1, 1]$

$y = \cos x$

D: $\mathbb{R}$

R: $[-1, 1]$

$y = \tan x$

D: $x \neq \frac{\pi}{2}n$: R: $\mathbb{R}$